

Lab: 13

Structures in C++

Objectives

In this lab we will learn:

- To group different types of data under one name.
- To store related information in a single unit.
- To create user-defined data types in C++.
- To make programs more organized and easy to manage.
- To represent real-world entities such as students, employees, books, etc.
- To simplify handling of large amounts of related data.
- To improve code readability and reusability.
- To allow passing multiple related values to functions together.

1: Structures in C++

A structure in C++ is a user-defined data type that allows us to combine different types of data items under a single name. It is used to store related information together in one unit.

- A structure is a collection of simple variables that have same or different data types:
- int, float, char and so on.
- This is unlike the array in which all the variables must be the same type.
- The data items in a structure are called the members of the structure.
- The closing brace in the structure type declaration must be followed by a semicolon.
- It is important to understand that a structure type declaration does not tell the compiler to reserve any space in memory.
- All a structure declaration does is, it defines the 'form' of the structure.
- One structure can be nested within another structure.
- Like an ordinary variable, a structure variable can also be passed to a function.
- We may either pass individual structure elements or the entire structure variable at one go.

2: Defining a Structure

To define a structure, we need to use the keyword struct. The general form of a structure definition in C++ is given as follows

Syntax:

```
struct StructureName
{
    dataType member1;
    dataType member2;
    dataType member3;
};
```

Example: Structure with one structure variable

```
#include
using namespace std;
struct part:      // declare a structure
{
    int modelnumber;    // ID number of car (structure member)
    int partnumber;    // ID number of car part (structure member)
    float cost;        // cost of part (structure member)
};
int main()
{
    part part1;        // define a structure variable
    // give values to structure members
    part1.modelnumber = 6244;
    part1.partnumber = 373;
    part1.cost = 217.55F;
    // display structure members
    cout << " Model : " << part1.modelnumber << endl;
    cout << " Part : " << part1.partnumber << endl;
    cout << " Cost : " << part1.cost << " $" << endl;
    return 0;
}
```

Output:

Model: 6244

Part: 373

Cost: 217.55 \$

Example: Structure with two structure variable

```
#include
using namespace std;
struct part {          // declare a structure
    int modelnumber;    // ID number of car (structure member)
    int partnumber;     // ID number of car part (member)
    float cost;         // cost of part (structure member)
};

int main() {
    part part1 = {6244, 373, 217.55F}; // initialize variable
    part part2;                        //declare a structure variable
    // display structure members related to first structure variable,
    cout << " Model : " << part1.modelnumber << endl; 1
    cout << " Part : " << part1.partnumber << endl; 1
    cout << " Cost : " << part1.cost << " $" << endl;
    part2 = part1;                    // assign 1st variable to 2nd (copying)
    // display members related to 2nd structure variable, part2
    cout << " Model : " << part2.modelnumber << endl;
    cout << " Part : " << part2.partnumber << endl;
    cout << " Cost : " << part2.cost << " $" << endl;
    return 0; }
```


Output:

Model: 6244

Part: 373

Cost: 217.55 \$

Model: 6244

Part: 373

Cost: 217.55 \$

Example: Employee Information using Structure

```
#include
using namespace std;
struct Person
{
    char name[50];
    int age;
    float salary;
};
int main() {
    Person p1;
    cout << "Enter Full name: ";
    cin.get(p1.name, 50);
    cout << "Enter age: ";
    cin >> p1.age;
    cout << "Enter salary: ";
```

```
cin >> p1.salary;  
  
cout << "\nDisplaying Information." << endl;  
  
cout << "Name: " << p1.name << endl;  
  
cout << "Age: " << p1.age << endl;  
  
cout << "Salary: " << p1.salary;  
  
return 0;}
```

Output:

Enter Full name: Murad Ali

Enter age: 34

Enter salary: 1200

Displaying Information.

Name: Murad Ali

Age: 34

Salary: 1200

Lab Tasks:

Task 1: Structure Declaration and Initialization

- Declare a structure called Person with the following members:
 - name (string)
 - age (integer)
 - address (string).
- Declare a variable called person1 of type Person and initialize its members with values of your choice. Display the values of the members of person1 using dot notation.

```
#include <iostream>           // for input/output
using namespace std;         // to use cout without std::
struct Person                 // structure declaration
{
    string name;              // member to store name
    int age;                  // member to store age
    string address;           // member to store address
}
int main(){
    Person person1;           // creating structure variable
    person1.name = "Asad";    // assigning name
    person1.age = 20;          // assigning age
    person1.address = "Peshawar"; // assigning address
    cout << "Name: " << person1.name ;    // display name
    cout << "Age: " << person1.age ;       // display age
    cout << "Address: " << person1.address ; // display address
    return 0; }               // program ends
```

Task 2: Structure Declaration and Initialization

- Declare a structure called Student with the following members:
 - name (string)– age (integer)
 - grade (char)
- Declare a variable of type Student called student1 and initialize its members with values of your choice. Display values of student1 structure members.

```
#include <iostream>
using namespace std;
struct Student      // structure declaration
{
    string name;      // student name
    int age;          // student age
    char grade;       // student grade
};
int main()
{
    Student student1; // structure variable
    student1.name = "Ali"; // assigning name
    student1.age = 19; // assigning age
    student1.grade = 'A'; // assigning grade
    cout << "Name: " << student1.name << endl; // display name
    cout << "Age: " << student1.age << endl; // display age
    cout << "Grade: " << student1.grade << endl; // display grade
    return 0;}

```


Task 3: Accessing Structure Elements

- Declare a structure called Book with the following members:
 - title (string)
 - author (string)
 - price (float)
 - pages (integer)
- Declare a variables of type Book called book1 and book2, and initialize its members with values of your choice.
- Display values of structure members of variables book1 and book2 .

```
#include <iostream>

using namespace std;

struct Book // structure declaration
{
    string title; // book title
    string author; // author name
    float price; // book price
    int pages; // total pages
};

int main()
{
    Book book1; // first book variable
    Book book2; // second book variable
```

```
book1.title = "C++ Basics";    // title of first book

book1.author = "Bjarne";       // author

book1.price = 500.5;           // price

book1.pages = 300;             // pages

book2.title = "Programming";   // title of second book

book2.author = "John";         // author

book2.price = 650;             // price

book2.pages = 450;             // pages

cout << "Book 1" << endl;

cout << book1.title << endl;

cout << book1.author << endl;

cout << book1.price << endl;

cout << book1.pages << endl;

cout << "Book 2" << endl;

cout << book2.title << endl;

cout << book2.author << endl;

cout << book2.price << endl;

cout << book2.pages << endl;

return 0;    // end

}
```